SRS: Data Retention

1. Overview

The data retention job deletes data from the data lake once it has passed its retention period, which depends on the storage layer. Data age is based on the `year=YYYY/month=MM` partition and measured in either calendar years or fiscal years. The job applies only to the buckets and prefixes in its configuration. It is scheduled monthly by Amazon EventBridge at the beginning of the month and runs as an AWS Lambda function. Backup, archiving, and legal-hold processes are outside the scope of this document.

2. Retention Policy

Calendar types

Type	Period
Calendar year	January to December (e.g. Jan 2026 to Dec 2026)
Fiscal year	April to March (e.g. Apr 2026 to Mar 2027)

Retention period by storage layer

Layer	Retention period
Bronze	3 years
Silver	3 years
Gold	6 years

Partition used for retention

Item	Rule
Partition format	`year=YYYY/month=MM` (e.g. `year=2026/month=04`)
Applies to	Bronze, silver, and gold
Age basis	The partition values, not the S3 last-modified date

3. Functional Requirements

Requirement	Rationale
The job must process only the buckets and prefixes in the configuration, each with its storage layer and calendar type (calendar year or fiscal year).	Limits deletion to approved locations.
The retention period per layer must be configurable without a code change.	Policy changes should not need a new release.

Requirement	Rationale
A partition is eligible for deletion once its calendar or fiscal year ended more than the retention period ago. Prefixes without a valid `year=/month=` partition must be skipped and logged.	Gives one clear rule and avoids deleting data that cannot be dated.
The job must run monthly at the beginning of the month, scheduled by EventBridge and executed by Lambda. It must delete only eligible partitions, and re-running it must be safe.	Keeps retention automatic and prevents accidental deletion.
Each run must log the partitions, objects, and size deleted per prefix, and alert on failure.	Provides an audit trail and makes failures visible.

4. Acceptance Criteria

- The job runs automatically at the beginning of each month through EventBridge and executes in Lambda.
- Only buckets and prefixes listed in the configuration are processed, and prefixes that do not match the `year=/month=` format are skipped, logged, and not deleted.
- Bronze and silver partitions older than 3 years and gold partitions older than 6 years are deleted, while newer partitions are kept.
- Calendar year and fiscal year datasets each get the correct deletion cutoff, and changing the retention period or calendar type in the configuration takes effect on the next run without a code change.
- Each run produces a log with the partitions, objects, and size deleted per prefix, and a failed run triggers an alert.